Penjelasan Koding Frontend — MiniProjectKPI_EWI

Dokumen ini menjelaskan isi file-file Dart di folder `frontend/lib/` dengan bahasa sederhana. Semua file di bawah ini **aktif dipakai** kecuali yang ditandai kosong.

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1. File: `main.dart` (88 baris)

File ini adalah **titik awal aplikasi** — menjalankan Flutter, mendaftarkan semua Provider, dan menentukan layar pertama.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`MyCustomScrollBehavior`	Custom scroll agar bisa drag pakai mouse/trackpad di web/desktop
2	def	`buildScrollbar(...)`	Tampilkan scrollbar yang terlihat dan bisa diklik
3	def	`main()`	**Entry point** — jalankan `ExpenseApp`
4	**Class**	`ExpenseApp`	Widget utama — daftarkan semua Provider dan atur tema
5	def	`build(context)`	Pasang MultiProvider (6 provider) → pilih Login atau Dashboard

Provider yang didaftarkan:

Provider	Sumber Data
`AuthProvider`	Token login + data user
`ThemeProvider`	Mode gelap/terang
`SettlementProvider`	Data settlement, expense, kategori
`AdvanceProvider`	Data kasbon
`RevenueProvider`	Data pemasukan
`TaxProvider`	Data pajak

Ringkas: 88 baris | 2 Class | 3 Function

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2. File: `services/api_service.dart` (855 baris) ★ FILE TERBESAR DI SERVICES

File ini adalah **penghubung Flutter ke server backend**. Semua HTTP request (GET, POST, PUT, DELETE) ada di sini.

Autentikasi

No	Tipe	Nama	Penjelasan Sederhana
1	def	`setToken(token)`	Simpan JWT token untuk request berikutnya
2	def	`login(username, password)`	POST login → dapat token
3	def	`getMe()`	GET profil user yang sedang login

CRUD Settlement

No	Tipe	Nama	Penjelasan Sederhana
4	def	`getSettlements({...})`	List settlement + filter
5	def	`getSettlement(id)`	Detail 1 settlement
6	def	`createSettlement(...)`	Buat settlement baru
7	def	`updateSettlement(id, {...})`	Edit settlement
8	def	`deleteSettlement(id)`	Hapus settlement
9	def	`submitSettlement(id)`	Submit untuk di-review
10	def	`approveAllSettlement(id)`	Manager approve semua expense
11	def	`rejectAllSettlement(id, {notes})`	Manager reject + alasan
12	def	`completeSettlement(id)`	Tandai selesai

CRUD Expense

No	Tipe	Nama	Penjelasan Sederhana
13	def	`createExpense({...})`	Tambah expense + upload file evidence (multipart)
14	def	`updateExpense(id, {...})`	Edit expense + ganti evidence
15	def	`deleteExpense(id)`	Hapus expense
16	def	`approveExpense(id, action, {notes})`	Manager approve/reject 1 expense

CRUD Kategori

No	Tipe	Nama	Penjelasan Sederhana
17	def	`getCategories()`	List kategori approved
18	def	`createCategory(name, {parentId})`	Buat kategori baru
19	def	`updateCategory(id, name)`	Edit kategori

No	Tipe	Nama	Penjelasan Sederhana
20	def	`deleteCategory(id)`	Hapus kategori
21	def	`getPendingCategories()`	List kategori pending
22	def	`approveCategory(id, action)`	Manager approve/reject kategori

CRUD Kasbon (Advance)

No	Tipe	Nama	Penjelasan Sederhana
23	def	`getAdvances({...})`	List kasbon + filter
24	def	`createAdvance(title, desc, {type})`	Buat kasbon baru
25	def	`getAdvance(id)`	Detail 1 kasbon
26	def	`updateAdvance(id, {...})`	Edit kasbon
27	def	`deleteAdvance(id)`	Hapus kasbon
28	def	`addAdvanceItem({...})`	Tambah item kasbon + evidence
29	def	`updateAdvanceItem(id, {...})`	Edit item kasbon
30	def	`deleteAdvanceItem(id)`	Hapus item kasbon
31	def	`submitAdvance(id)`	Submit kasbon
32	def	`approveAllAdvance(id, {notes})`	Manager approve kasbon
33	def	`rejectAllAdvance(id, {notes})`	Manager reject kasbon

Export Laporan

No	Tipe	Nama	Penjelasan Sederhana
34	def	`exportAdvanceExcel({...})`	Download Excel kasbon
35	def	`getBulkSettlementsPdf({...})`	Download PDF banyak settlement
36	def	`getBulkAdvancesPdf({...})`	Download PDF banyak kasbon
37	def	`getAdvanceReceipt(id)`	Download PDF 1 kasbon
38	def	`exportExcel({...})`	Download Excel summary expense
39	def	`getSummary({...})`	Ambil data summary JSON
40	def	`getSummaryPdf({...})`	Download PDF summary
41	def	`getReceipt(id)`	Download PDF kuitansi settlement
42	def	`getEvidenceUrl(filename)`	URL gambar evidence

Settings & Annual Report

No	Tipe	Nama	Penjelasan Sederhana
43	def	`getStorageSettings()`	Ambil setting folder storage
44	def	`updateStorageSettings(dir)`	Ubah folder storage
45	def	`getRevenues({...})`	List revenue

No	Tipe	Nama	Penjelasan Sederhana
46	def	`createRevenue(data)`	Buat revenue
47	def	`updateRevenue(id, data)`	Edit revenue
48	def	`deleteRevenue(id)`	Hapus revenue
49	def	`getTaxes({...})`	List pajak
50	def	`createTax(data)`	Buat pajak
51	def	`updateTax(id, data)`	Edit pajak
52	def	`deleteTax(id)`	Hapus pajak
53	def	`getAnnualReport({year, refresh})`	Ambil data laporan tahunan JSON
54	def	`getReportYearSettings()`	Ambil tahun default laporan
55	def	`updateReportYearSettings(year)`	Ubah tahun default
56	def	`getAnnualReportPdf({year, refresh})`	Download PDF laporan tahunan
57	def	`getAnnualReportExcel({year, refresh})`	Download Excel laporan tahunan
58	def	`_handleResponse(res)`	*(internal)* — Parse response HTTP, lempar error jika gagal

Ringkas: 855 baris | 1 Class | 58 Function

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3. File: `theme/app_theme.dart` (243 baris)

Mendefinisikan **tema visual** (warna, font, spacing) untuk mode gelap dan terang.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`AppTheme`	Kumpulan semua warna, style, dan ThemeData
2	Variable	`primary`, `primaryDark`, `accent`	Warna-warna brand utama
3	Variable	`success`, `warning`, `danger`	Warna status (hijau, kuning, merah)
4	Variable	`background`, `surface`, `card`	Warna latar belakang
5	Variable	`lightTheme`	ThemeData untuk mode terang
6	Variable	`darkTheme`	ThemeData untuk mode gelap

Ringkas: 243 baris | 1 Class | 0 Function (semua static variable/getter)

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4. File: `utils/currency_formatter.dart` (39 baris)

Format input angka agar otomatis pakai separator ribuan Indonesia (titik).

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`CurrencyInputFormatter`	TextInputFormatter khusus format angka Indonesia
2	def	`formatEditUpdate(old, new)`	Saat user ketik angka → otomatis format "1.000.000"

Ringkas: 39 baris | 1 Class | 1 Function

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5. File: `utils/file_helper.dart` (171 baris)

Helper untuk **menyimpan file export** di folder lokal dan membukanya.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`FileHelper`	Kumpulan fungsi static untuk simpan/buka file
2	def	`_resolveExportDirectory()`	Tentukan folder export (Windows: `data/`, Android: external storage)
3	def	`saveAndOpenFolder({...})`	Simpan file + buka folder
4	def	`saveAndOpenFile({...})`	Simpan file + langsung buka file-nya
5	def	`saveFile({...})`	⭐ **Fungsi utama** — simpan bytes ke file, tampilkan snackbar sukses/gagal
6	def	`_resolveWritableFilePath(path)`	Hindari overwrite — tambahkan timestamp jika file sudah ada
7	def	`_ensureTimestampedFilename(name)`	Pastikan nama file punya timestamp
8	def	`openExportFolder()`	Buka folder export di file explorer
9	def	`openFile(path)`	Buka file dengan aplikasi default OS
10	def	`formatTimestamp([dateTime])`	Format "20260305_2140"

Ringkas: 171 baris | 1 Class | 9 Function

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6. File: `utils/responsive_layout.dart` (40 baris)

Widget untuk **layout responsif** — otomatis ganti tampilan sesuai ukuran layar.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`ResponsiveLayout`	Widget yang pilih layout berdasarkan lebar layar
2	def	`isMobile(context)`	Cek lebar < 700px
3	def	`isTablet(context)`	Cek lebar 700–1100px
4	def	`isDesktop(context)`	Cek lebar ≥ 1100px
5	def	`build(context)`	Tampilkan widget `mobile`, `tablet`, atau `desktop`

Ringkas: 40 baris | 1 Class | 4 Function

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7. File: `providers/auth_provider.dart` (94 baris)

State management login — simpan token, data user, dan status login.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`AuthProvider`	Kelola autentikasi + profil user
2	getter	`isLoggedIn`, `isManager`, `isStaff`, `isMitraEks`	Cek status user
3	getter	`fullName`, `role`, `roleDisplayName`	Info user
4	def	`_loadToken()`	*(private)* — Cek token tersimpan di SharedPreferences, refresh profil dari API
5	def	`login(username, password)`	Login → simpan token ke SharedPreferences
6	def	`logout()`	Hapus token + data user

Ringkas: 94 baris | 1 Class | 3 Function

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8. File: `providers/settlement_provider.dart` (428 baris) ★ PROVIDER TERBESAR

State management settlement, expense, dan kategori — penghubung antara UI dan ApiService.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`SettlementProvider`	Kelola semua data settlement + expense + kategori

CRUD Settlement

No	Tipe	Nama	Penjelasan Sederhana
2	def	`updateToken(token)`	Update token ke ApiService
3	def	`loadSettlements({...})`	Ambil list settlement dari API
4	def	`loadSettlement(id)`	Ambil detail 1 settlement
5	def	`createSettlement(title, desc, {advanceId, type})`	Buat settlement baru
6	def	`updateSettlement(id, {...})`	Edit settlement
7	def	`deleteSettlement(id)`	Hapus settlement
8	def	`submitSettlement(id)`	Submit settlement
9	def	`completeSettlement(id)`	Selesaikan settlement

CRUD Expense

No	Tipe	Nama	Penjelasan Sederhana
10	def	`addExpense({...})`	Tambah expense + upload evidence
11	def	`updateExpense({...})`	Edit expense
12	def	`deleteExpense(id, settlementId)`	Hapus expense
13	def	`approveExpense(id, action, {notes})`	Approve/reject expense
14	def	`approveAllSettlement(id)`	Approve semua expense
15	def	`rejectAllSettlement(id, {notes})`	Reject semua expense

CRUD Kategori

No	Tipe	Nama	Penjelasan Sederhana
16	def	`loadCategories()`	Ambil list kategori
17	def	`createCategory(name, {parentId})`	Buat kategori
18	def	`updateCategory(id, name)`	Edit kategori
19	def	`deleteCategory(id)`	Hapus kategori
20	def	`loadPendingCategories()`	Ambil kategori pending
21	def	`approveCategory(id, action)`	Approve/reject kategori

Laporan

No	Tipe	Nama	Penjelasan Sederhana
22	def	`getEvidenceUrl(filename)`	URL gambar evidence
23	def	`exportExcel({...})`	Download Excel summary
24	def	`getSummary({...})`	Data summary JSON
25	def	`getSummaryPdf({...})`	Download PDF summary
26	def	`getReceipt(id)`	Download PDF kuitansi
27	def	`getBulkPdf({...})`	Download PDF bulk settlement

Ringkas: 428 baris | 1 Class | 26 Function

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9. File: `providers/advance_provider.dart` (233 baris)

State management kasbon (advance) — CRUD header, item, workflow, dan export.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`AdvanceProvider`	Kelola semua data kasbon
2	def	`updateToken(token)`	Update token
3	def	`loadAdvances({...})`	Ambil list kasbon + filter
4	def	`loadAdvance(id)`	Detail 1 kasbon
5	def	`createAdvance(title, desc, {type})`	Buat kasbon
6	def	`updateAdvance(id, title, desc, {type})`	Edit kasbon
7	def	`deleteAdvance(id)`	Hapus kasbon
8	def	`addAdvanceItem(advId, catId, desc, amount, {file})`	Tambah item kasbon
9	def	`updateAdvanceItem(itemId, catId, desc, amount, {file})`	Edit item
10	def	`deleteAdvanceItem(itemId)`	Hapus item
11	def	`submitAdvance(id)`	Submit kasbon
12	def	`approveAdvance(id, {notes})`	Manager approve
13	def	`rejectAdvance(id, notes)`	Manager reject
14	def	`getAdvanceReceipt(id)`	Download PDF kuitansi
15	def	`exportExcel({...})`	Download Excel kasbon

No	Tipe	Nama	Penjelasan Sederhana
16	def	`getBulkPdf({...})`	Download PDF bulk kasbon

Ringkas: 233 baris | 1 Class | 15 Function

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10. File: `providers/revenue_provider.dart` (98 baris)

State management revenue (pemasukan) — CRUD sederhana.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`RevenueProvider`	Kelola data revenue
2	def	`fetchRevenues({startDate, endDate})`	Ambil list revenue + filter tanggal
3	def	`createRevenue(data, {startDate, endDate})`	Buat revenue → auto reload
4	def	`updateRevenue(id, data, {startDate, endDate})`	Edit revenue → auto reload
5	def	`deleteRevenue(id, {startDate, endDate})`	Hapus revenue → auto reload

Ringkas: 98 baris | 1 Class | 4 Function

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11. File: `providers/tax_provider.dart` (94 baris)

State management pajak (tax) — CRUD sederhana, sama pola dengan `RevenueProvider`.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`TaxProvider`	Kelola data pajak
2	def	`fetchTaxes({startDate, endDate})`	Ambil list pajak + filter tanggal
3	def	`createTax(data, {startDate, endDate})`	Buat pajak → auto reload
4	def	`updateTax(id, data, {startDate, endDate})`	Edit pajak → auto reload
5	def	`deleteTax(id, {startDate, endDate})`	Hapus pajak → auto reload

Ringkas: 94 baris | 1 Class | 4 Function

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12. File: `providers/theme_provider.dart` (45 baris)

Pengaturan tema gelap/terang — simpan pilihan di SharedPreferences.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`ThemeProvider`	Kelola mode tema (dark/light/system)
2	def	`_loadThemeMode()`	Baca tema tersimpan saat init
3	def	`setThemeMode(mode)`	Ubah tema + simpan
4	def	`_parseThemeMode(mode)`	Konversi string → ThemeMode

Ringkas: 45 baris | 1 Class | 3 Function

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13. File: `screens/login_screen.dart` (218 baris)

Halaman **login** — tampilan kartu tengah dengan animasi fade-in.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`LoginScreen`	Widget StatefulWidget halaman login
2	**Class**	`_LoginScreenState`	State: animasi, controller input, logic login
3	def	`initState()`	Setup animasi fade-in
4	def	`dispose()`	Bersihkan controller
5	def	`_login()`	Proses login via AuthProvider
6	def	`build(context)`	UI: card login (username, password, tombol masuk, hint default user)

Ringkas: 218 baris | 2 Class | 4 Function

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14. File: `screens/dashboard_screen.dart` (1953 baris) ★ FILE TERBESAR

Halaman utama setelah login — **sidebar navigasi + konten dinamis** (settlement list, kasbon, laporan, kategori, settings).

Widget Utama

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`DashboardScreen`	Layout utama: sidebar + konten berdasarkan page index
2	**Class**	`_Sidebar`	**Sidebar navigasi** — menu item, info user, tombol logout
3	**Class**	`_NavItem`	Item navigasi di sidebar (dengan hover effect)

No	Tipe	Nama	Penjelasan Sederhana
4	**Class**	`_PageSelector`	Dropdown pemilih halaman (Apple-style) untuk mobile
5	def	`_formatNumber(num)`	Format angka ribuan

Settlement List View (Halaman 0)

No	Tipe	Nama	Penjelasan Sederhana
6	**Class**	`_SettlementListView`	Halaman list settlement utama
7	def	`_loadDefaultReportYearAndReload()`	Ambil tahun default → reload data
8	def	`_pickDateRange(context)`	Dialog pilih rentang tanggal
9	def	`_clearDateRange()`	Reset filter tanggal
10	def	`_reloadSettlements()`	Reload data settlement
11	def	`_exportExcel()`	Download Excel summary
12	def	`_exportPdf()`	Download PDF bulk settlement
13	def	`build(context)`	UI: toolbar filter + grid settlement cards
14	def	`_showCreateDialog(context)`	Dialog buat settlement baru (pilih tipe, link advance)

Widget Pendukung

No	Tipe	Nama	Penjelasan Sederhana
15	**Class**	`_FilterChip`	Chip filter status (draft/submitted/approved/dll)
16	**Class**	`_SettlementCard`	Kartu settlement (title, status badge, total, tanggal)
17	**Class**	`_StatusBadge`	Badge warna status

Manajemen Kategori (Manager, Halaman 3)

No	Tipe	Nama	Penjelasan Sederhana
18	**Class**	`_CategoryManagementView`	Halaman kelola kategori (tree view)
19	def	`_buildCategoryTile(cat)`	Render 1 kategori + sub-kategori
20	def	`_showAddCategoryDialog(context)`	Dialog tambah kategori (bisa pilih parent)
21	def	`_showEditCategoryDialog(context, cat)`	Dialog edit nama kategori
22	def	`_deleteCategory(id)`	Konfirmasi + hapus kategori
23	def	`_approveCategory(id, action)`	Approve/reject kategori pending

Ringkas: 1953 baris | 10 Class | ~30 Function

15. File: `screens/settlement_detail_screen.dart` (1745 baris) ⭐ FILE #2 TERBESAR

Halaman **detail 1 settlement** — lihat expense, tambah/edit/hapus expense, upload evidence, submit, approve/reject.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`SettlementDetailScreen`	Widget utama detail settlement
2	def	`_statusColor(status)`	Warna berdasarkan status (draft=abu, approved=hijau, dll)
3	def	`_toDouble(value)`	Konversi dynamic → double
4	def	`_formatExpenseAmount(expense)`	Format jumlah IDR + info mata uang asing
5	def	`build(context)`	Loading / Error / Content
6	def	`_buildContent(context, s, auth, prov)`	⭐ **UI utama** — summary cards + tabel expense + action buttons
7	def	`_buildActionCell(exp, auth, s)`	Tombol approve/reject/edit/hapus per expense
8	def	`_showAddExpenseDialog(context)`	Dialog buat expense (pilih kategori pohon, upload file, amount, mata uang)
9	def	`_showEvidence(path, filename)`	Dialog lihat gambar evidence
10	def	`_approveExpense(id, action)`	Approve/reject 1 expense
11	def	`_deleteExpense(id)`	Hapus 1 expense
12	def	`_submitSettlement()`	Submit settlement
13	def	`_completeSettlement()`	Selesaikan settlement
14	def	`_downloadReceipt()`	Download PDF kuitansi
15	def	`_exportSettlementExcel()`	Download Excel settlement
16	def	`_showEditSettlementDialog(context, s)`	Dialog edit title/desc settlement
17	def	`_showEditExpenseDialog(context, exp)`	Dialog edit expense (sama UI dengan add)
18	def	`_showAddCategoryDialog(ctx, {...})`	Dialog buat kategori baru langsung dari form expense
19	**Class**	`_SummaryCard`	Kartu ringkasan (Total, Approved, Pending)
20	**Class**	`_ActionButton`	Tombol aksi dengan icon +

No	Tipe	Nama	Penjelasan Sederhana
			gradient

Ringkas: 1745 baris | 3 Class | ~18 Function

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16. File: `screens/report_screen.dart` (504 baris)

Halaman **laporan ringkasan expense** — tabel per kategori per bulan + export Excel/PDF.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`ReportScreen`	Halaman laporan summary
2	def	`_loadSummary()`	Ambil data summary dari API
3	def	`_pickDateRange()`	Filter rentang tanggal
4	def	`_clearDateRange()`	Reset filter
5	def	`build(context)`	Layout: header + actions + tabel
6	def	`_buildHeaderInfo()`	Info tahun / rentang tanggal aktif
7	def	`_buildActions()`	Tombol export Excel, PDF, ke laporan tahunan
8	def	`_buildSummaryTable()`	⭐ **Tabel utama** — baris per kategori, kolom per bulan
9	def	`_exportFullExcel()`	Download Excel → simpan file
10	def	`_exportSummaryPdf()`	Download PDF → simpan file

Ringkas: 504 baris | 1 Class | 10 Function

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17. File: `screens/annual_report_screen.dart` (979 baris) ★

Halaman **laporan tahunan** — tampilkan data revenue, tax, expense + tabel detail + export Excel/PDF.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`AnnualReportScreen`	Halaman laporan tahunan
2	def	`_loadDefaultReportYearAndFetch()`	Ambil tahun default → fetch data
3	def	`_fetchReport({refresh})`	Ambil data annual dari API
4	def	`_exportPdf()`	Download PDF annual → simpan
5	def	`_exportExcel()`	Download Excel annual → simpan

Helper Format & Parse

No	Tipe	Nama	Penjelasan Sederhana
6	def	`_asListMap(data)`	Cast dynamic → List
7	def	`_toDouble(value)`	Cast dynamic → double
8	def	`_fmtNumber(value)`	Format angka dengan separator
9	def	`_fmtMoney(value)`	Format mata uang IDR/USD
10	def	`_fmtDate(value)`	Format tanggal "dd MMM yyyy"
11	def	`_parseDate(value)`	Parse string → DateTime
12	def	`_extractImportedRow(text)`	Baca "Imported from row X"
13	def	`_extractImportedSheetRow(text)`	Baca "Imported from Sheet1 row X"
14	def	`_extractBatchNumber(text)`	Ambil nomor batch dari title
15	def	`_isBatchSettlement(item)`	Cek apakah settlement tipe batch
16	def	`_cleanSettlementTitle(title)`	Buang prefix "Single:", "Batch:"
17	def	`_groupAnnualExpenses(expenses)`	Kelompokkan expense per settlement
18	def	`_expenseCategoryIndex(name)`	Mapping nama kategori → index kolom

Builder UI

No	Tipe	Nama	Penjelasan Sederhana
19	def	`_isNumericHeader(header)`	Cek apakah kolom numerik
20	def	`_isCenterColumn(header, index)`	Cek apakah kolom di-center
21	def	`_columnAlignment(header, index)`	Tentukan alignment kolom
22	def	`_buildTableCard({title, headers, rows, ...})`	Render 1 tabel dalam card
23	def	`_buildCacheInfo()`	Info cache (tanggal generate)
24	def	`_buildInputButtons()`	Tombol ke halaman Revenue & Tax
25	def	`_buildDisplayTables()`	⭐ **Semua tabel** (Revenue, Tax, Expense single, batch)
26	def	`build(context)`	Layout utama: summary + tabel + export
27	def	`_buildSummaryCards()`	3 kartu ringkasan (Revenue, Cost, Profit)
28	def	`_buildCard(title, amount, color)`	1 kartu summary

Ringkas: 979 baris | 1 Class | 28 Function

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18. File: `screens/settings_screen.dart` (428 baris)

Halaman **pengaturan** — tema, folder storage, tahun laporan.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`SettingsScreen`	Halaman settings
2	def	`_loadSettings()`	Ambil setting storage + tahun dari API
3	def	`_saveSettings()`	Simpan folder storage baru ke API
4	def	`_saveReportYearSettings()`	Simpan tahun default baru ke API
5	def	`_themeLabel(mode)`	Label: “Gelap”, “Terang”, “Sistem”
6	def	`_buildThemeSettingsCard(provider)`	Card pilihan tema (3 radio button)
7	def	`_buildStorageSettingsCard()`	Card input folder storage
8	def	`_buildReportYearSettingsCard()`	Card input tahun laporan
9	def	`build(context)`	Layout 3 kartu setting

Ringkas: 428 baris | 1 Class | 9 Function

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19. File: `screens/revenue_management_screen.dart` (463 baris)

Halaman **kelola data revenue** (pemasukan) — tabel data + dialog tambah/edit/hapus.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`RevenueManagementScreen`	Halaman CRUD revenue
2	def	`_loadData()`	Ambil data revenue dari API
3	def	`_saveRevenue({payload, id})`	Simpan/update revenue
4	def	`_toDouble(value)`	Parse angka dari string input
5	def	`_showRevenueDialog([data])`	⭐ Dialog form revenue (10+ field: tanggal, klien, nilai, mata uang, PPN, PPh, dll)
6	def	`pickDate(ctrl)`	Date picker dialog
7	def	`_deleteRevenue(item)`	Konfirmasi + hapus revenue
8	def	`_textField(ctrl, label)`	Builder field teks
9	def	`_numberField(ctrl, label)`	Builder field angka
10	def	`_dateField(ctrl, label, pickDate)`	Builder field tanggal
11	def	`build(context)`	Tabel data revenue + tombol tambah

Ringkas: 463 baris | 1 Class | 10 Function

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20. File: `screens/tax_management_screen.dart` (418 baris)

Halaman **kelola data pajak** — sama pola dengan Revenue Management.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`TaxManagementScreen`	Halaman CRUD pajak
2	def	`_loadData()`	Ambil data pajak
3	def	`_toDouble(value)`	Parse angka
4	def	`_saveTax({payload, id})`	Simpan/update pajak
5	def	`_showTaxDialog([data])`	Dialog form pajak (tanggal, deskripsi, nilai, PPN, PPh 21/23/26)
6	def	`pickDate(ctrl)`	Date picker
7	def	`_deleteTax(item)`	Konfirmasi + hapus pajak
8	def	`_textField(ctrl, label)`	Builder field teks
9	def	`_numberField(ctrl, label)`	Builder field angka
10	def	`_dateField(ctrl, label, pickDate)`	Builder field tanggal
11	def	`build(context)`	Tabel data pajak + tombol tambah

Ringkas: 418 baris | 1 Class | 10 Function

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21. File: `screens/manager/manager_dashboard_screen.dart` (231 baris)

Halaman **dashboard khusus manager** — lihat settlement + kasbon yang butuh approval.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`ManagerDashboardScreen`	Tab bar: Settlement vs Kasbon menunggu approval
2	def	`build(context)`	Layout tab bar + tab view
3	def	`_buildSettlementsForApproval()`	List settlement status “submitted” → klik buka detail
4	def	`_buildAdvancesForApproval()`	List kasbon status “submitted” → klik buka dialog
5	def	`_showAdvanceApprovalDialog(advance, provider)`	Dialog approve/reject kasbon

Ringkas: 231 baris | 1 Class | 4 Function

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22. File: `screens/manager/manager_settlement_detail_screen.dart` (439 baris)

Halaman **detail settlement untuk manager** — review expense satu per satu atau sekaligus.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`ManagerSettlementDetailScreen`	Detail settlement view untuk manager
2	def	`_toDouble(value)`	Konversi dynamic → double
3	def	`_formatExpenseAmount(expense)`	Format jumlah expense (handle mata uang asing)
4	def	`build(context)`	Layout: info + expense list + tombol aksi
5	def	`_buildSettlementInfo(settlement)`	Card info settlement (title, status, creator)
6	def	`_buildExpensesList(expenses, provider)`	List expense + tombol approve/reject per item
7	def	`_buildApprovalActions(id, provider)`	Tombol “Approve All” dan “Reject All”
8	def	`_getStatusColor(status)`	Warna status
9	def	`_getIconForStatus(status)`	Icon status
10	def	`_approveExpense(id, provider)`	Approve 1 expense
11	def	`_rejectExpense(id, provider)`	Reject 1 expense + dialog alasan
12	def	`_approveAll(id, provider)`	Konfirmasi + approve semua
13	def	`_rejectAll(id, provider)`	Konfirmasi + reject semua + alasan

Ringkas: 439 baris | 1 Class | 12 Function

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23. File: `screens/advance/my_advances_screen.dart` (916 baris)

Halaman **list kasbon** — tampilan mirip settlement list (filter, buat baru, export).

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`MyAdvancesScreen`	List kasbon + filter + export
2	def	`_pickDateRange(context)`	Dialog pilih rentang tanggal
3	def	`_clearDateRange()`	Reset filter
4	def	`_reloadAdvances()`	Reload data kasbon
5	def	`_loadDefaultReportYearAndReload()`	Ambil tahun default → reload
6	def	`_exportExcel()`	Download Excel kasbon
7	def	`_exportPdf()`	Download PDF bulk kasbon
8	def	`_showCreateDialog(context)`	Dialog buat kasbon baru (title, desc, tipe single/batch)
9	def	`build(context)`	UI: toolbar filter + grid advance cards
10	**Class**	`_FilterChip`	Chip filter status
11	**Class**	`_AdvanceCard`	Kartu kasbon (title, status, total, tanggal)
12	**Class**	`_StatusBadge`	Badge warna status
13	def	`_formatNumber(value)`	Format angka ribuan

Ringkas: 916 baris | 4 Class | ~13 Function

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24. File: `screens/advance/advance_detail_screen.dart` (844 baris)

Halaman **detail 1 kasbon** — lihat item, tambah/edit/hapus item, submit, approve/reject, export.

No	Tipe	Nama	Penjelasan Sederhana
1	**Class**	`AdvanceDetailScreen`	Detail kasbon
2	def	`_statusColor(status)`	Warna sesuai status
3	def	`_displayAdvanceTitle(advance)`	Bersihkan prefix title
4	def	`_submitAdvance()`	Submit kasbon
5	def	`_approveAdvance()`	Manager approve
6	def	`_rejectAdvance()`	Manager reject + dialog alasan
7	def	`_downloadReceipt()`	Download PDF kuitansi kasbon
8	def	`_exportAdvanceExcel()`	Download Excel kasbon
9	def	`_showEditAdvanceDialog(context,`	Dialog edit title/desc/tipe kasbon

No	Tipe	Nama	Penjelasan Sederhana
		`advance)`	
10	def	`_showAddItemDialog(context, [item])`	Dialog tambah/edit item (kategori, amount, evidence)
11	def	`build(context)`	Loading / Error / Content
12	def	`_buildContent(context, adv, auth, prov)`	⭐ **UI utama** — summary + tabel item + tombol aksi
13	**Class**	`_SummaryCard`	Kartu ringkasan kasbon

Ringkas: 844 baris | 2 Class | 12 Function

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25. File: `screens/advance/advance_request_screen.dart` (0 baris) ❌ KOSONG

File ini **kosong** — tidak dipakai. Kemungkinan placeholder yang belum diimplementasi.

Ringkasan Total Frontend

File	Baris	Class	Function	Fungsi Utama
`main.dart`	88	2	3	Entry point + provider registration
`services/api_service.dart`	855	1	58	⭐ Semua HTTP request ke backend
`theme/app_theme.dart`	243	1	0	Tema gelap/terang
`utils/currency_formatter.dart`	39	1	1	Format angka Indonesia
`utils/file_helper.dart`	171	1	9	Simpan/buka file export
`utils/responsive_layout.dart`	40	1	4	Responsive mobile/tablet/desktop

File	Baris	Class	Function	Fungsi Utama
`providers/auth_provider.dart`	94	1	3	Login & profil user
`providers/settlement_provider.dart`	428	1	26	⭐ State settlement + expense
`providers/advance_provider.dart`	233	1	15	State kasbon
`providers/revenue_provider.dart`	98	1	4	State revenue
`providers/tax_provider.dart`	94	1	4	State pajak
`providers/theme_provider.dart`	45	1	3	State tema
`screens/login_screen.dart`	218	2	4	Halaman login
`screens/dashboard_screen.dart`	1953	10	30	⭐ Dashboard + sidebar + settlement list
`screens/settlement_detail_screen.dart`	1745	3	18	⭐ Detail settlement + CRUD expense
`screens/report_screen.dart`	504	1	10	Laporan ringkasan expense
`screens/annual_report_screen.dart`	979	1	28	⭐ Laporan tahunan

File	Baris	Class	Function	Fungsi Utama
`screens/settings_screen.dart`	428	1	9	Pengaturan tema/storage/tahun
`screens/revenue_management_screen.dart`	463	1	10	CRUD revenue
`screens/tax_management_screen.dart`	418	1	10	CRUD pajak
`screens/manager/manager_dashboard_screen.dart`	231	1	4	Dashboard approval manager
`screens/manager/manager_settlement_detail_screen.dart`	439	1	12	Review settlement (manager)
`screens/advance/my_advances_screen.dart`	916	4	13	List kasbon + filter
`screens/advance/advance_detail_screen.dart`	844	2	12	Detail kasbon + CRUD item
`screens/advance/advance_request_screen.dart`	0	0	0	❌ Kosong
TOTAL	**10576**	**41**	**~290**	

Catatan Penting

1. **File terbesar:** `dashboard_screen.dart` (1953 baris) dan `settlement_detail_screen.dart` (1745 baris) — keduanya mengandung banyak dialog dan widget helper
2. **File terpenting:** `api_service.dart` (58 function) — **semua komunikasi ke backend** lewat sini
3. **Pattern arsitektur:** Provider (ChangeNotifier) — setiap domain punya provider sendiri
4. **State management:** `SettlementProvider` paling besar (26 function) karena handle settlement + expense + kategori sekaligus
5. **Tema:** Menggunakan Google Fonts + mode gelap/terang yang tersimpan di SharedPreferences
6. **Responsive:** Mendukung mobile (<700px), tablet (700–1100px), dan desktop (≥1100px) via `ResponsiveLayout`
7. **File kosong:** `advance_request_screen.dart` — belum diimplementasi